University Of The South Pacific

CS219 Cloud Computing Assignment 2, 2024

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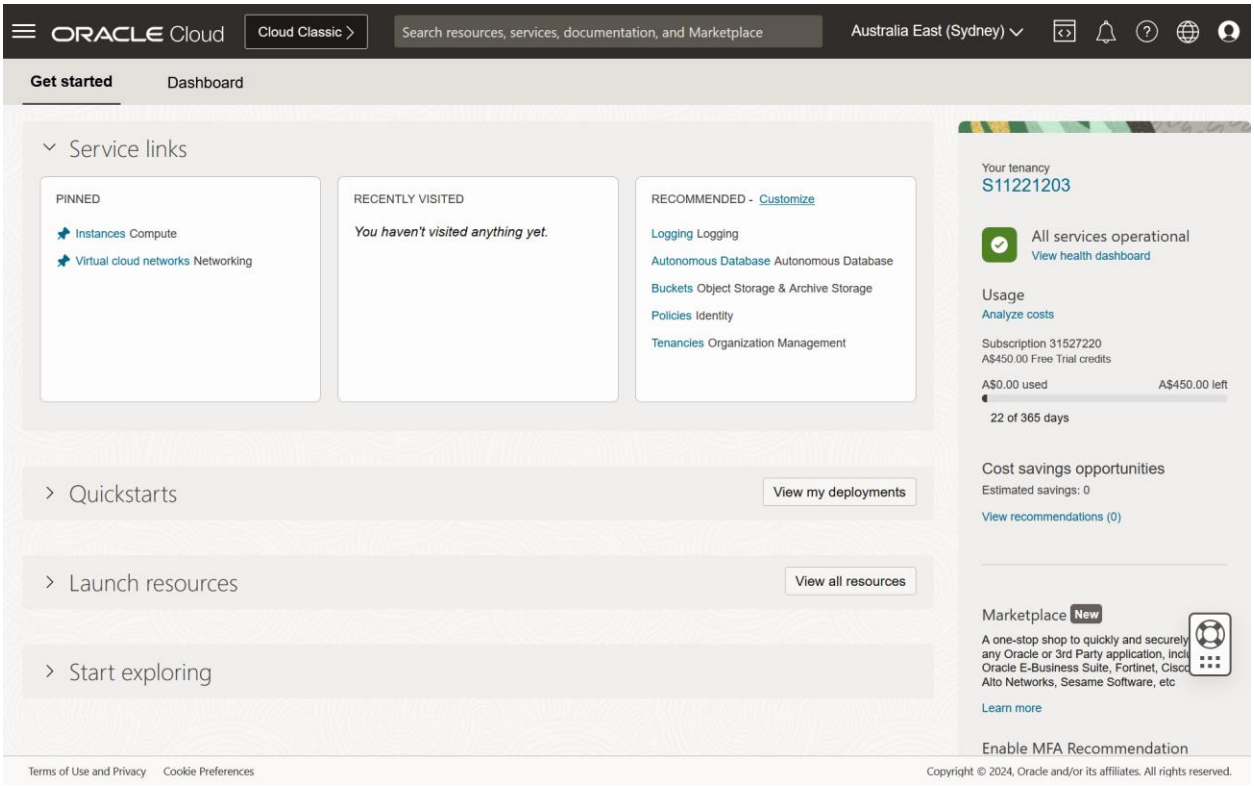
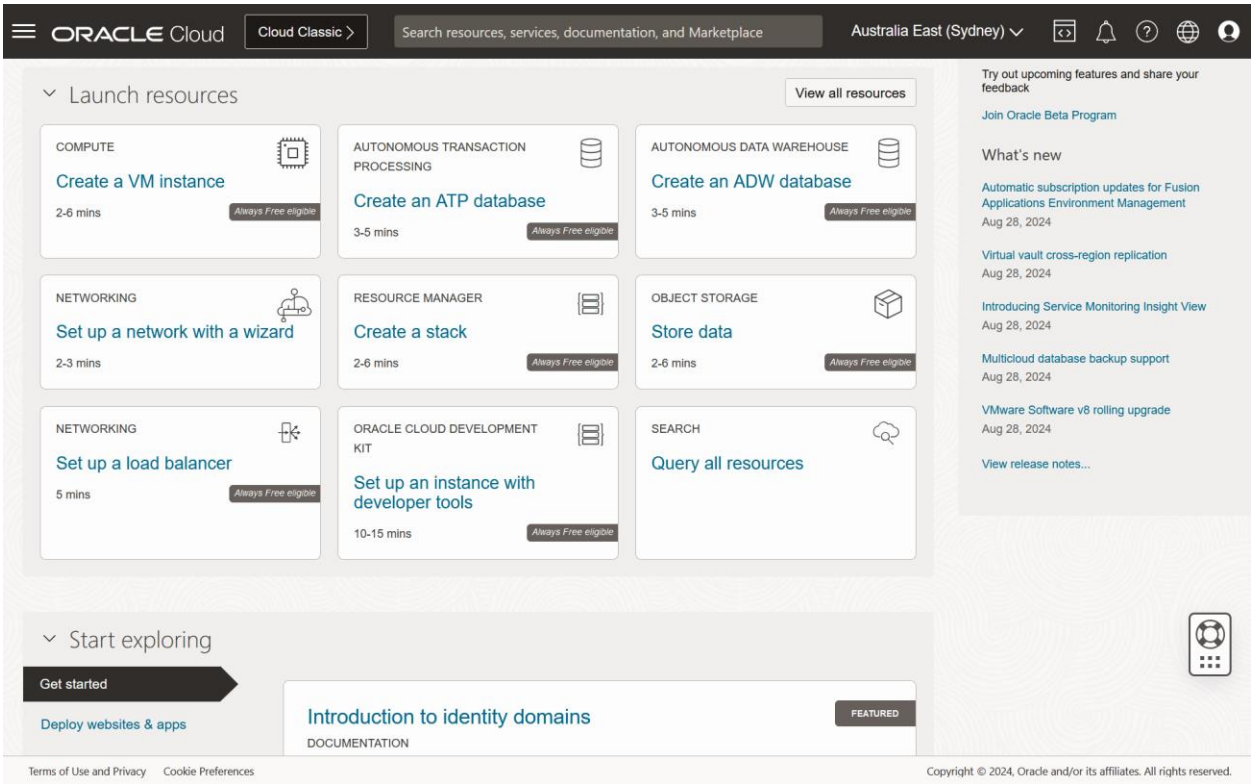
Question 1: Configure Ubuntu VM

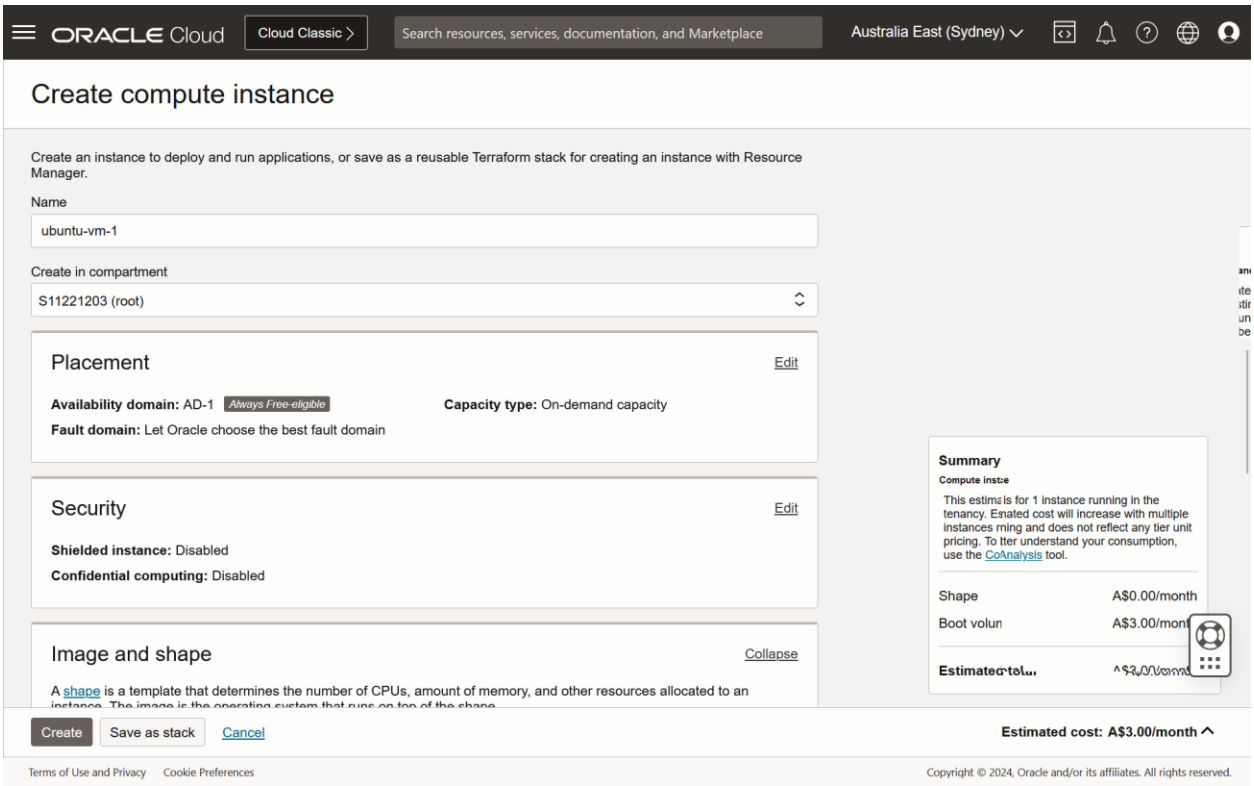
Step 1: Open a browser and log into https://cloud.oracle.com.



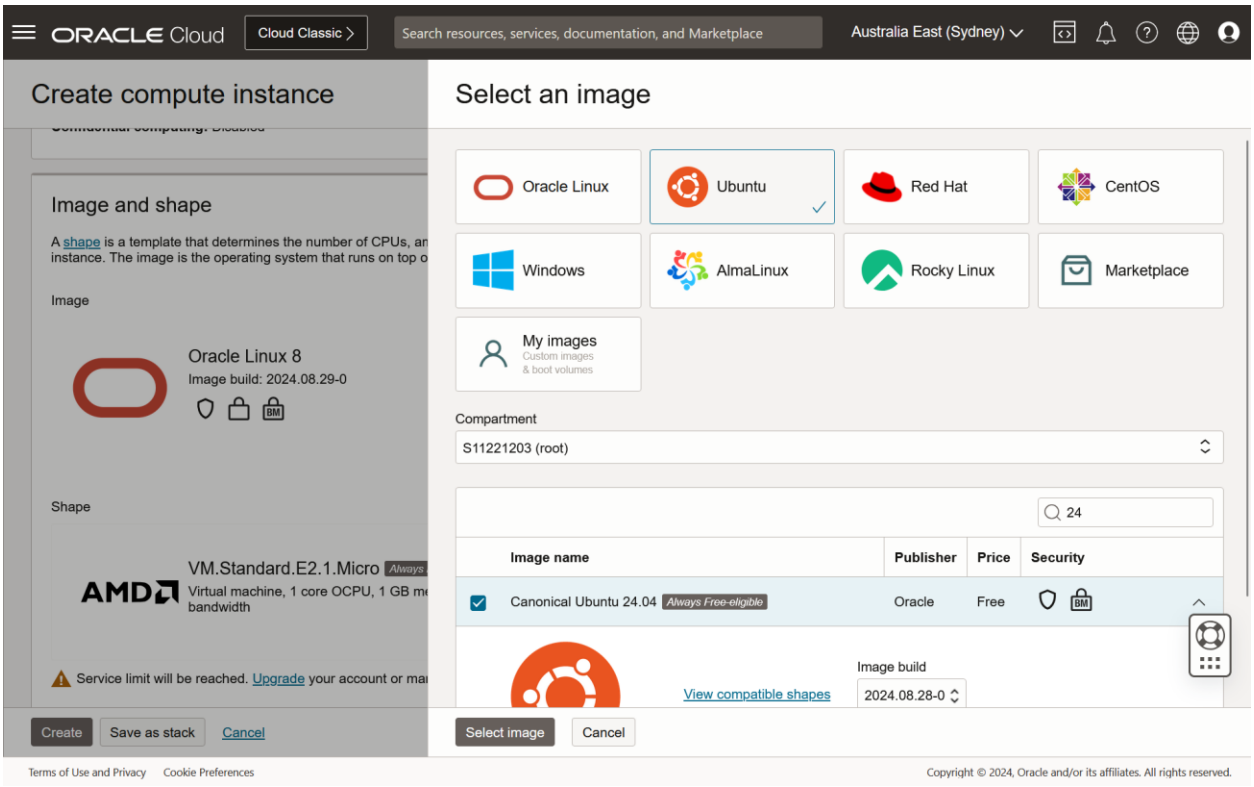
Step 2: After getting redirect to the Oracle Cloud homepage, scroll down to the “Launch Resources” section and click “Create a VM Instance”.



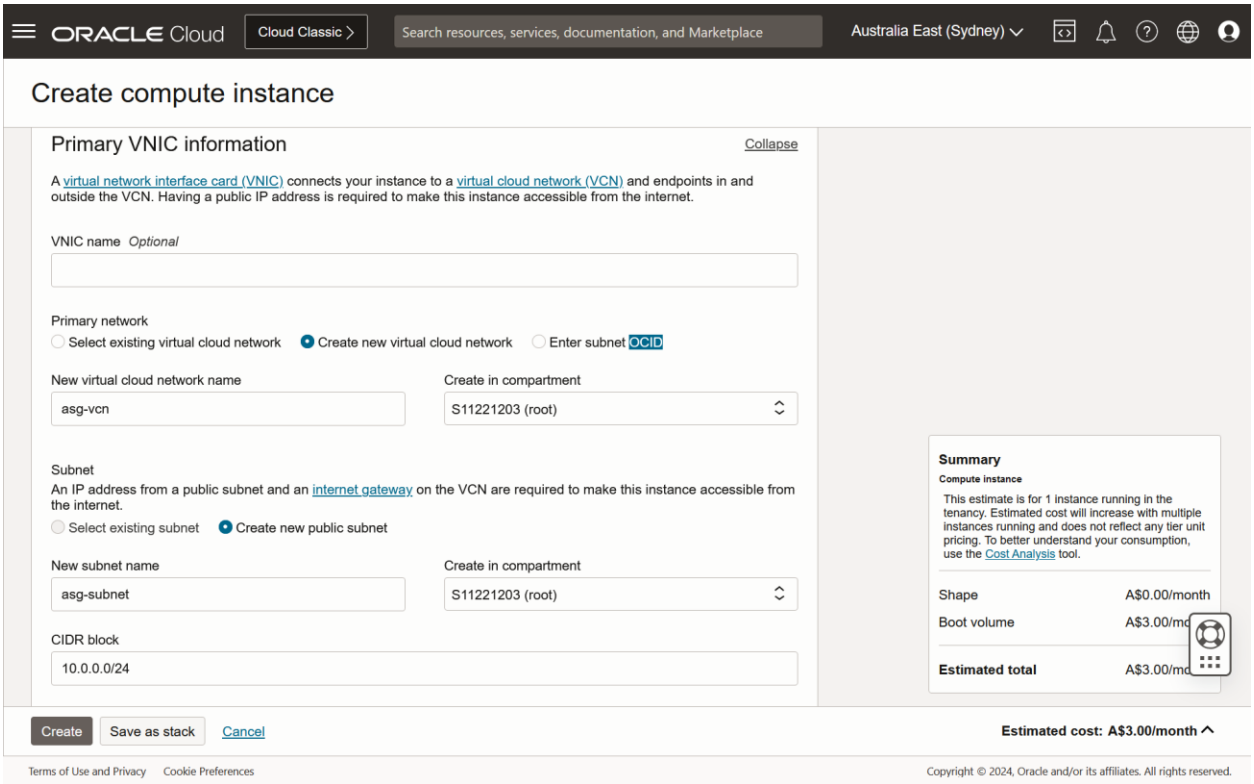
Step 3: Provide a name (like ubuntu-vm-1) and compartment (like S11221203) for the VM.



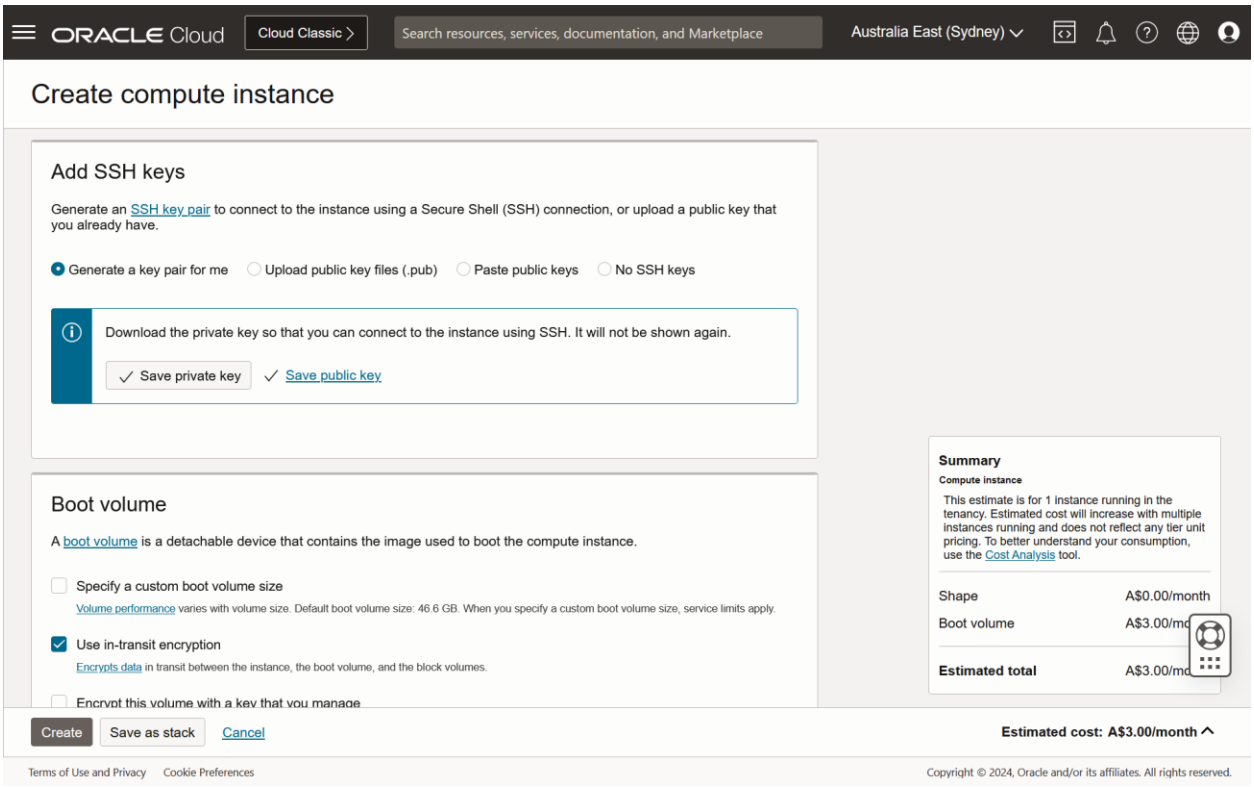
Step 4: Under the “Image and Shape” section, click “Edit”. Then click “Change Image” and select “Ubuntu”. Scroll down on this newly opened sidebar and select an Ubuntu version (like Canonical Ubuntu 24.04). Then press “Select Image” to confirm.



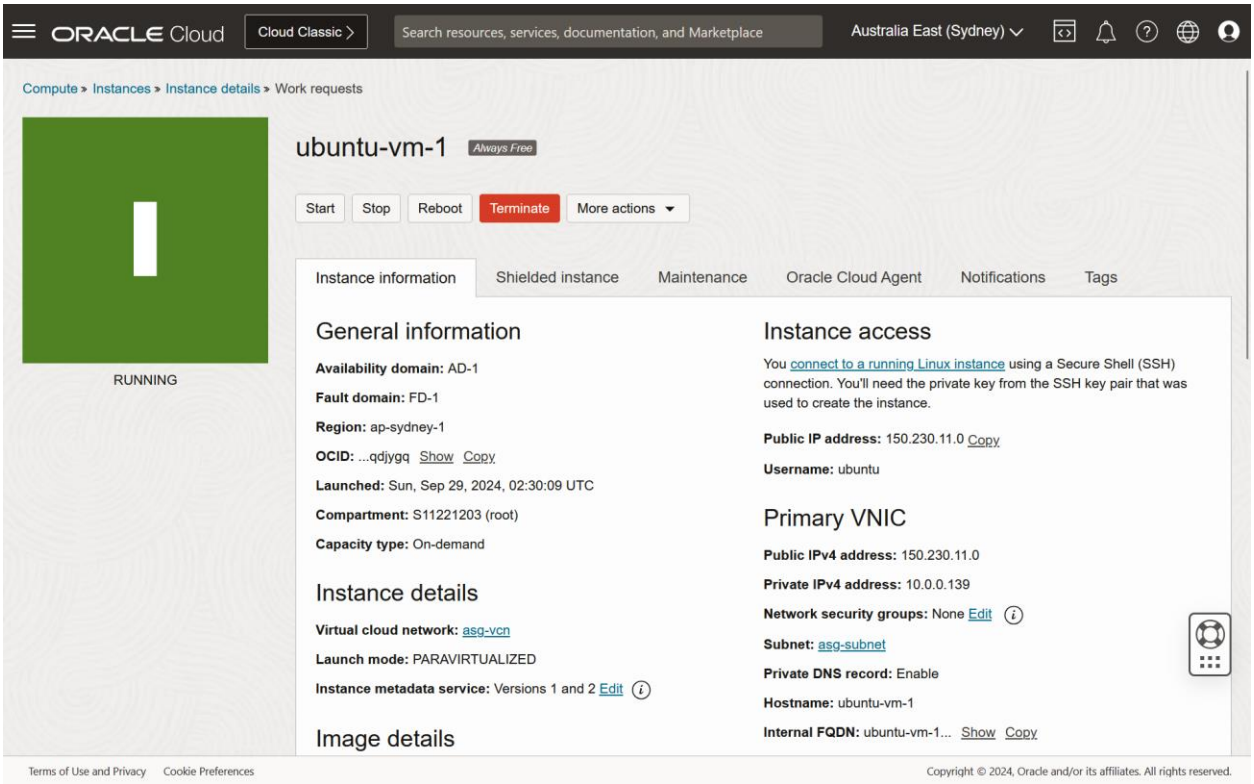
Step 5: : Under the “Primary VNIC Information” section, click “Edit”. In these settings under “Primary network”, select “Create new virtual cloud network” and give it a name (like asg-vcn). Similarly, under “Subnet”, select “Create new public subnet” and give it name (like asg-subnet).



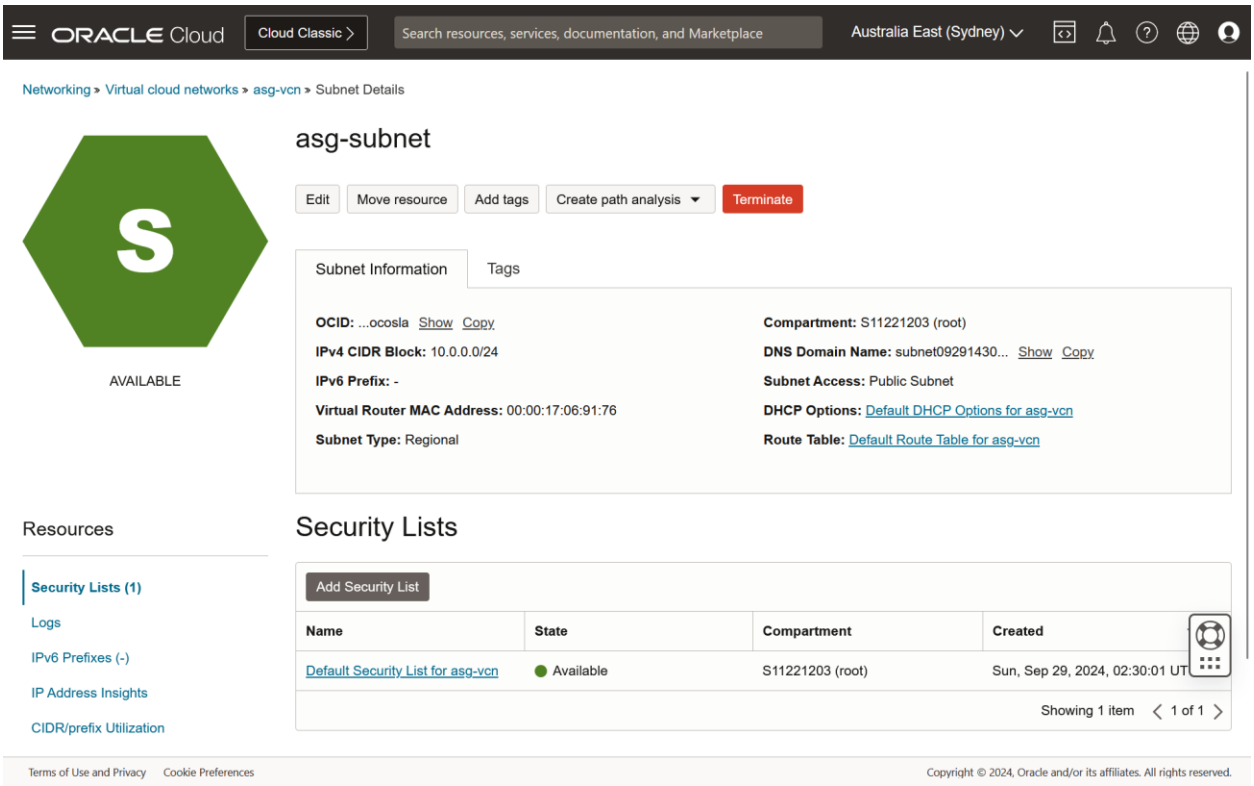
Step 6: Under the “Add SSH Keys” section, select “Generate a key pair for me”. Then, download both the public and private keys but clicking on “Save public key” and “Save private key”. Finally, click on the “Create” button on the bottom left corner.



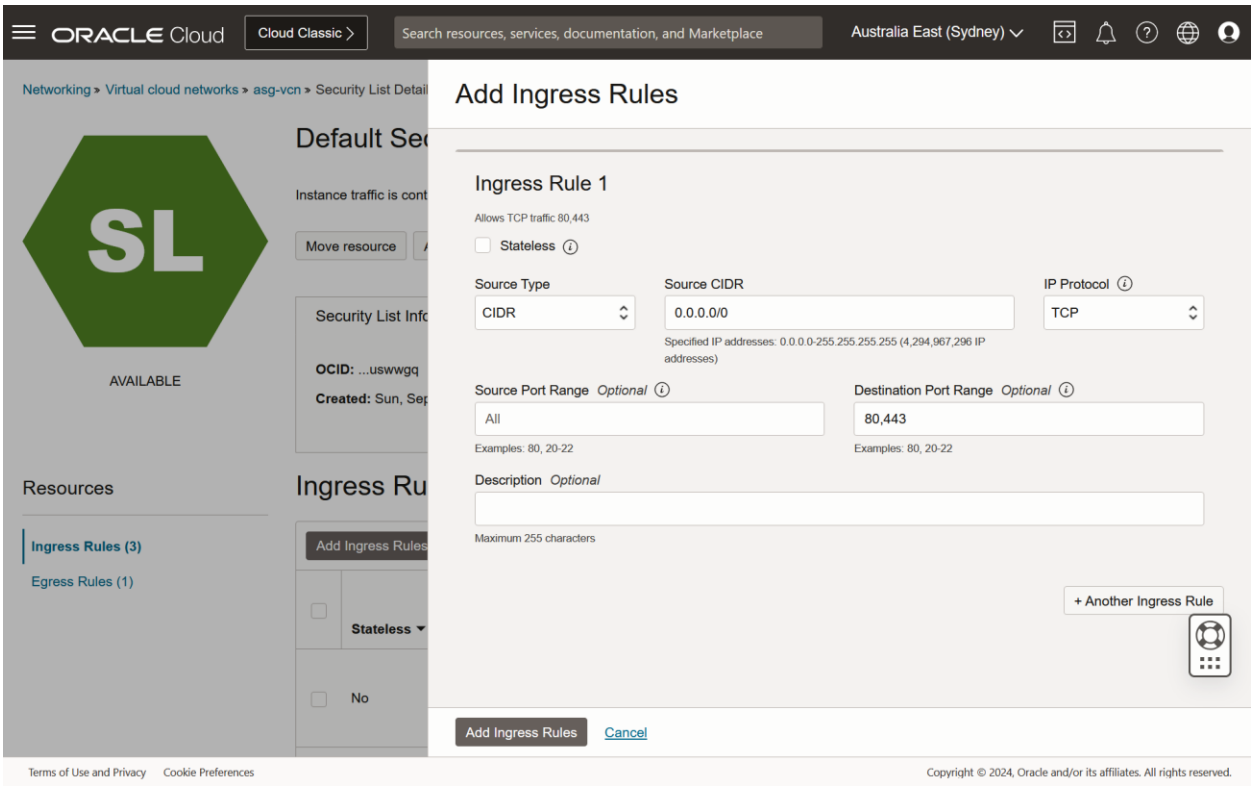
Step 7: After the Ubuntu VM has been created and is running, click on the subnet name (asg-subnet) that was created earlier under the “Primary VNIC” details.



Step 8: Click on “Default Security List for asg-vcn” at the bottom of the page.



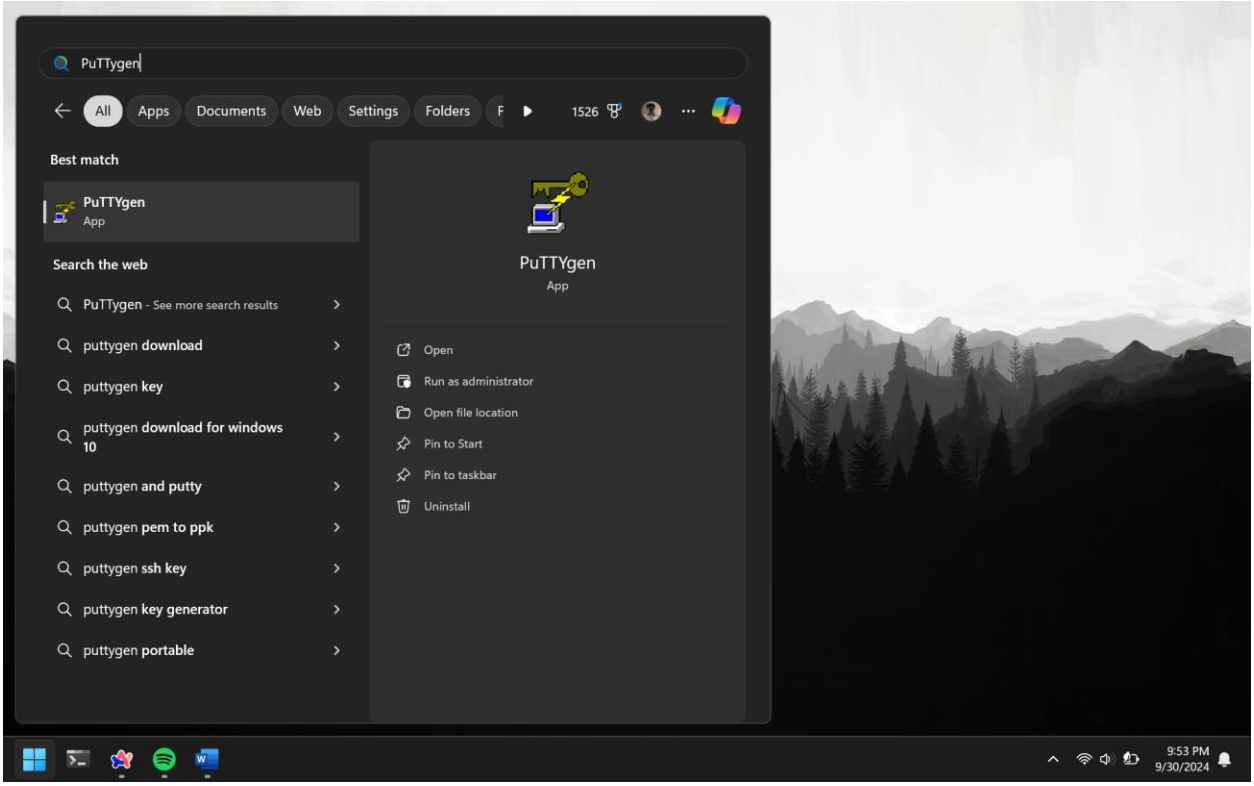
Step 9: Click on “Add ingress rule”. Set the source CIDR to “0.0.0.0/0” and the destination port range to “80,443” to enable internet access via HTTP and HTTPS. Click “Add Ingress Rules” to confirm.



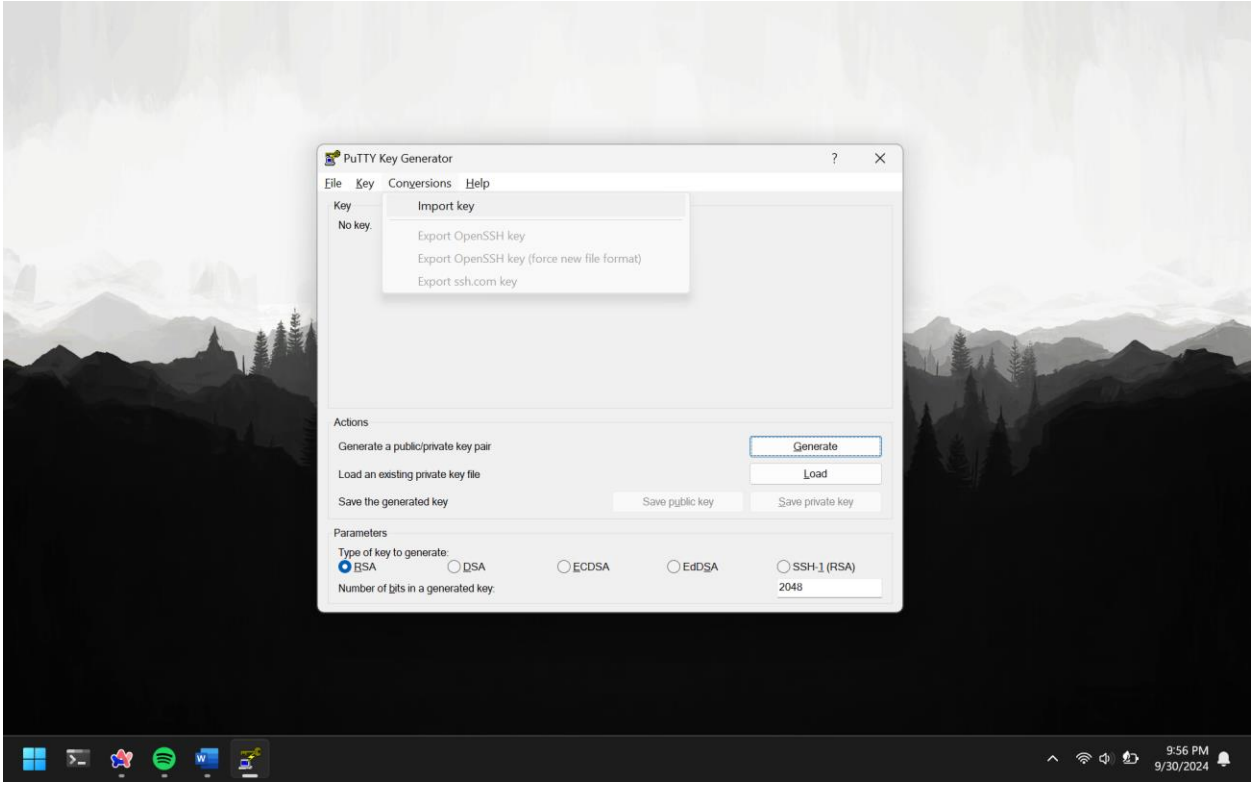
Question 1: Access VM Using PuTTY

Step 1: Download and install PuTTY. PuTTYgen comes included with PuTTY.

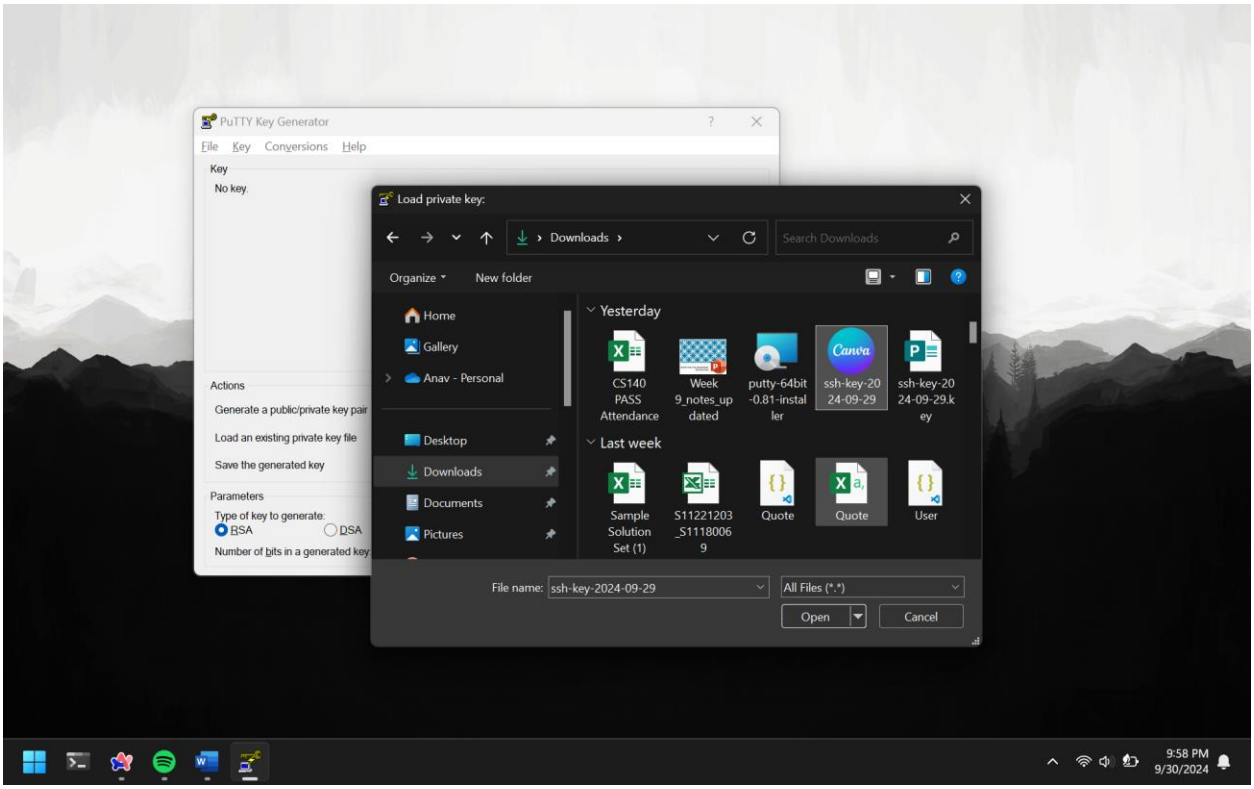
Step 2: Search for the “PuTTYgen” application. Click to open it.



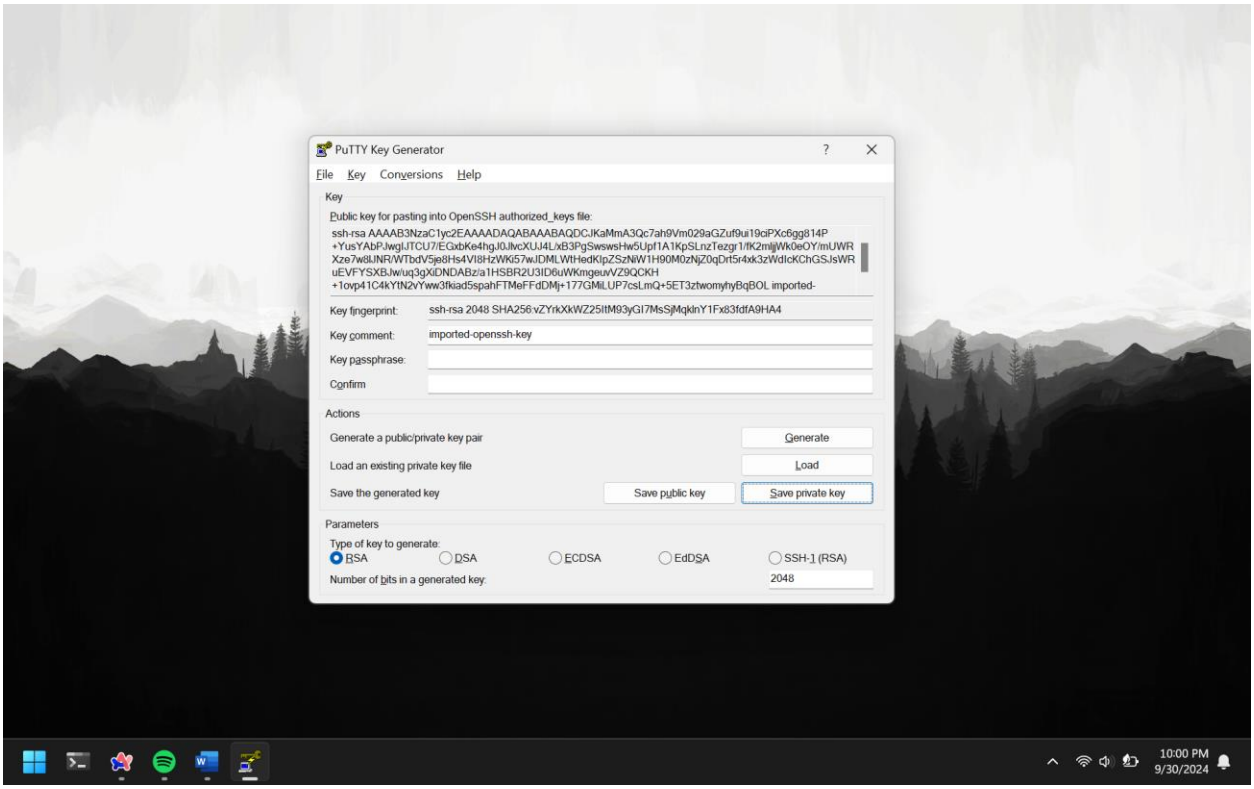
Step 3: In the top menu, click on “conversions” and select “Import Key”.



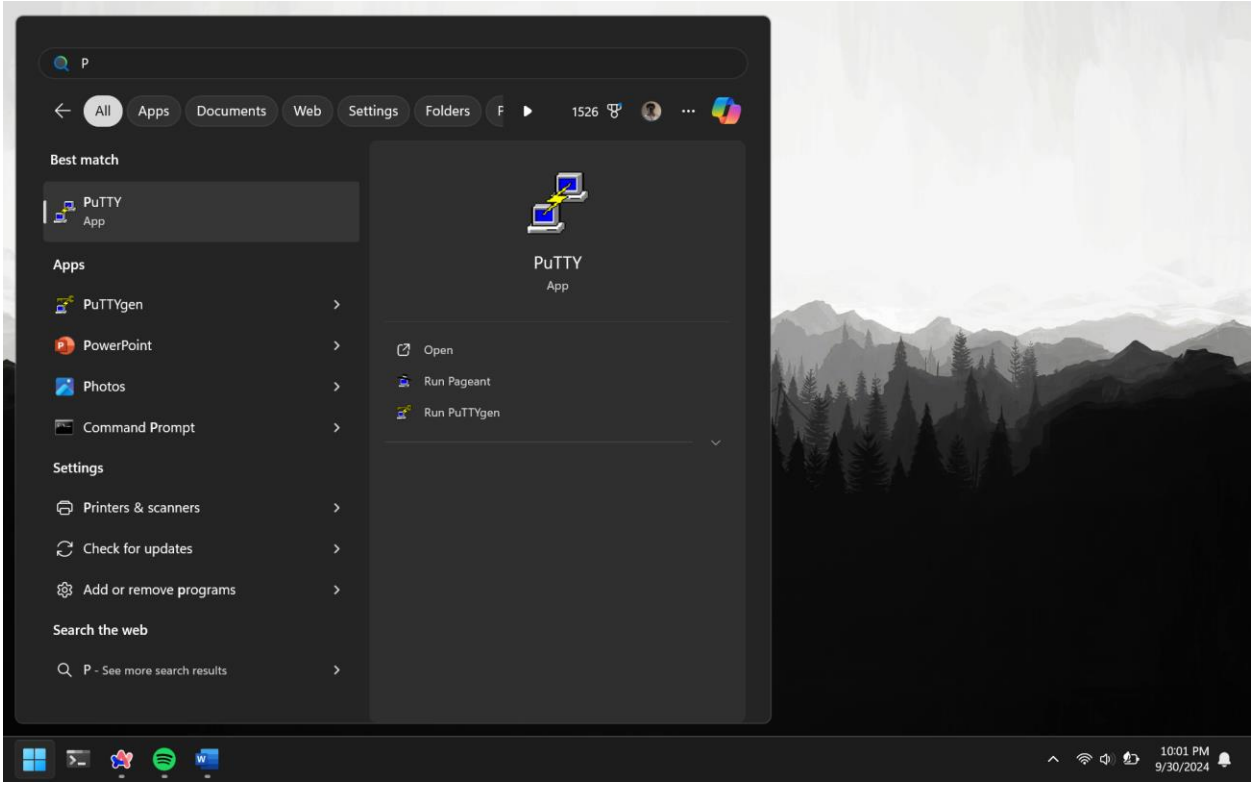
Step 4: Select the private key that was previously downloaded during the Ubuntu VM configuration process.



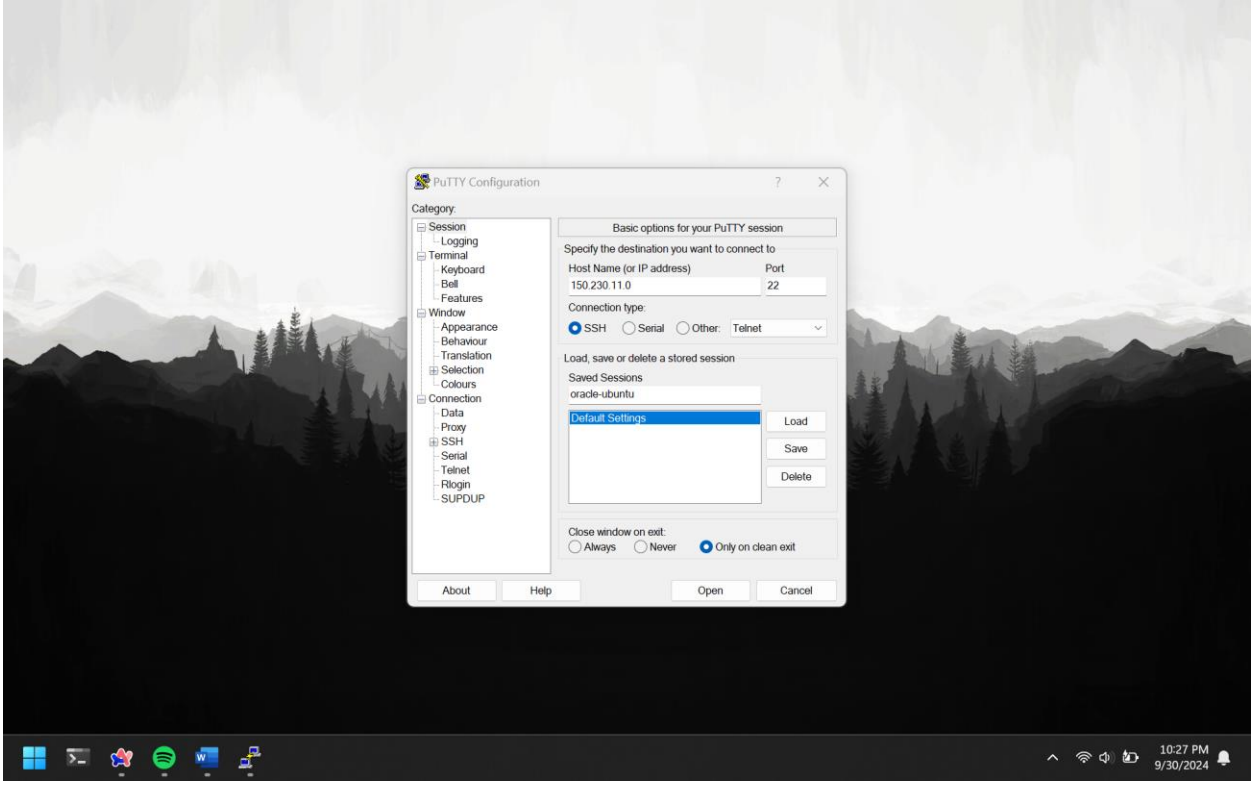
Step 5: Click on “Save private key” to save the new version of the private key. This new version is what PuTTY will use to connect to the VM.



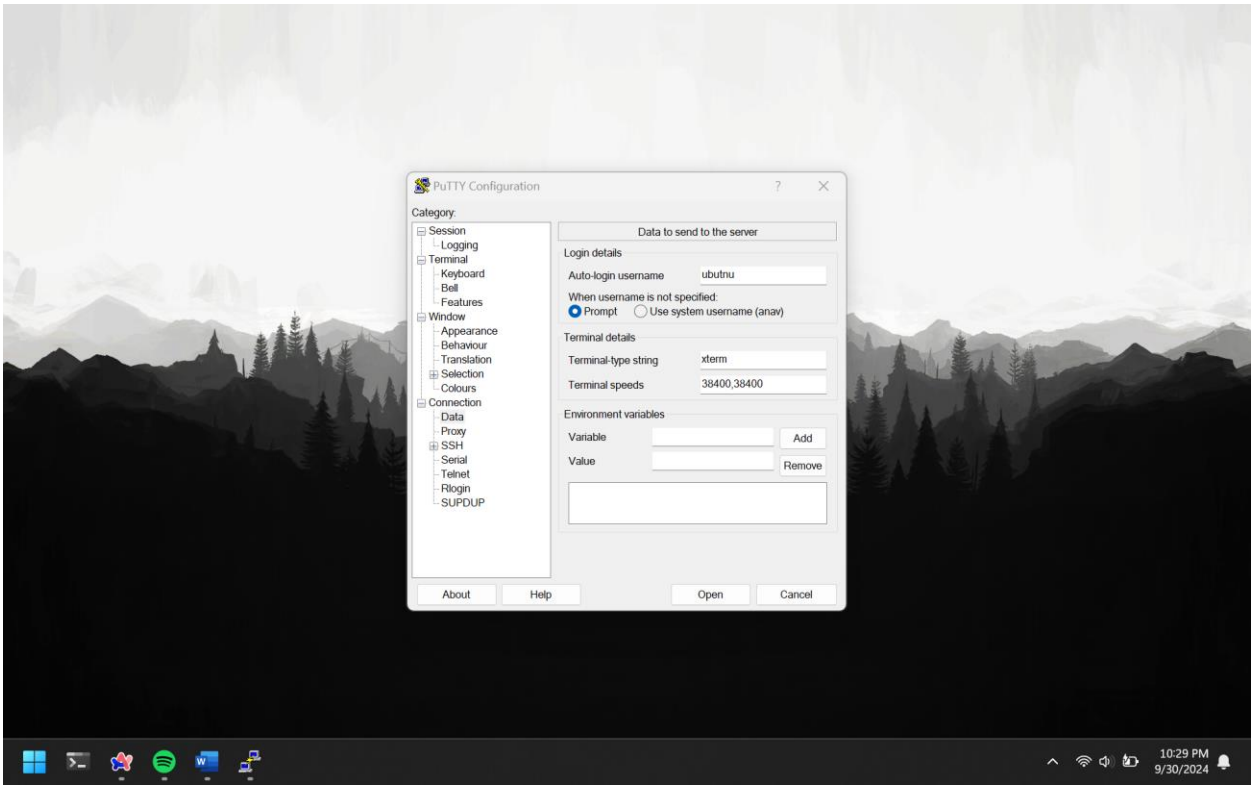
Step 6: Search for the “PuTTY” application. Click to open it.



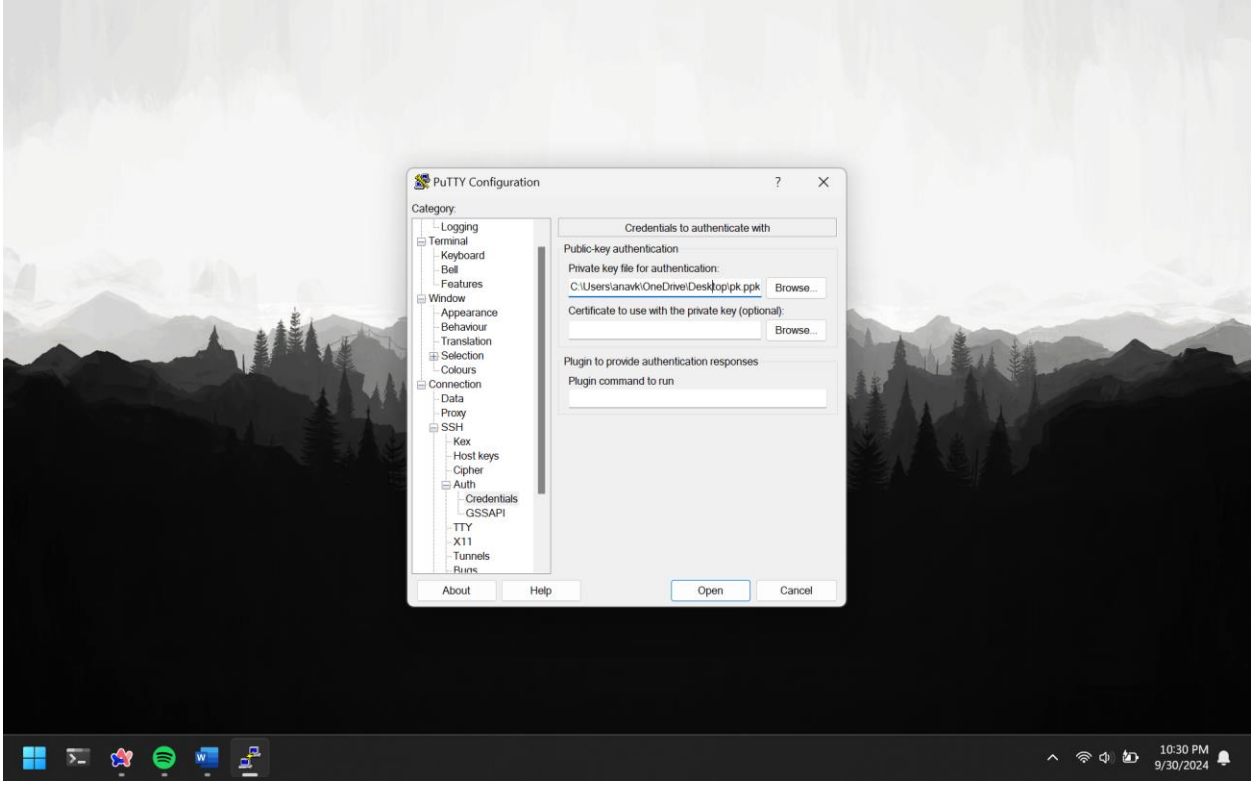
Step 7: Enter the public IP address of the VM and provide a session name. The public IP address can be found on Oracle Cloud. Go to https://cloud.oracle.com/dashboard, click on “instances” and select the desired VM to get its details.



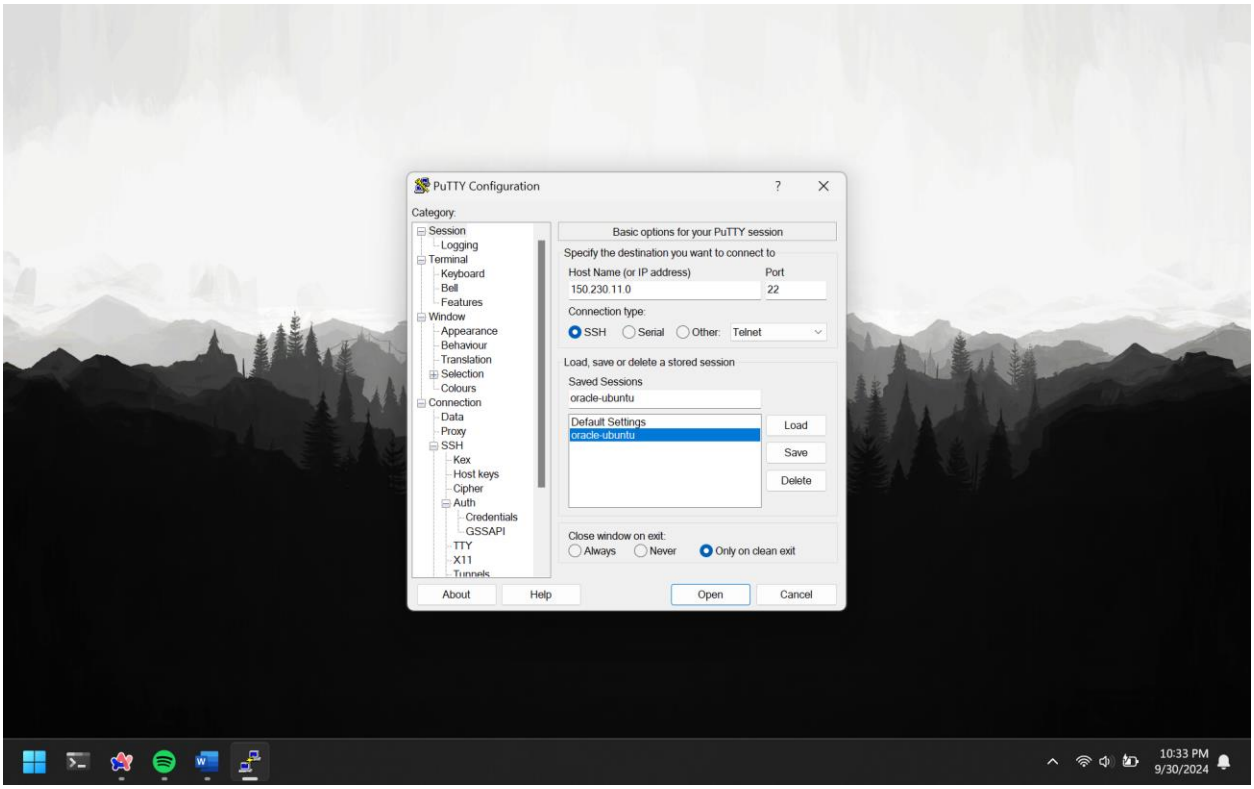
Step 8: On the left sidebar, click on “connection” and then choose “data”. Enter “ubuntu” as the auto login name.



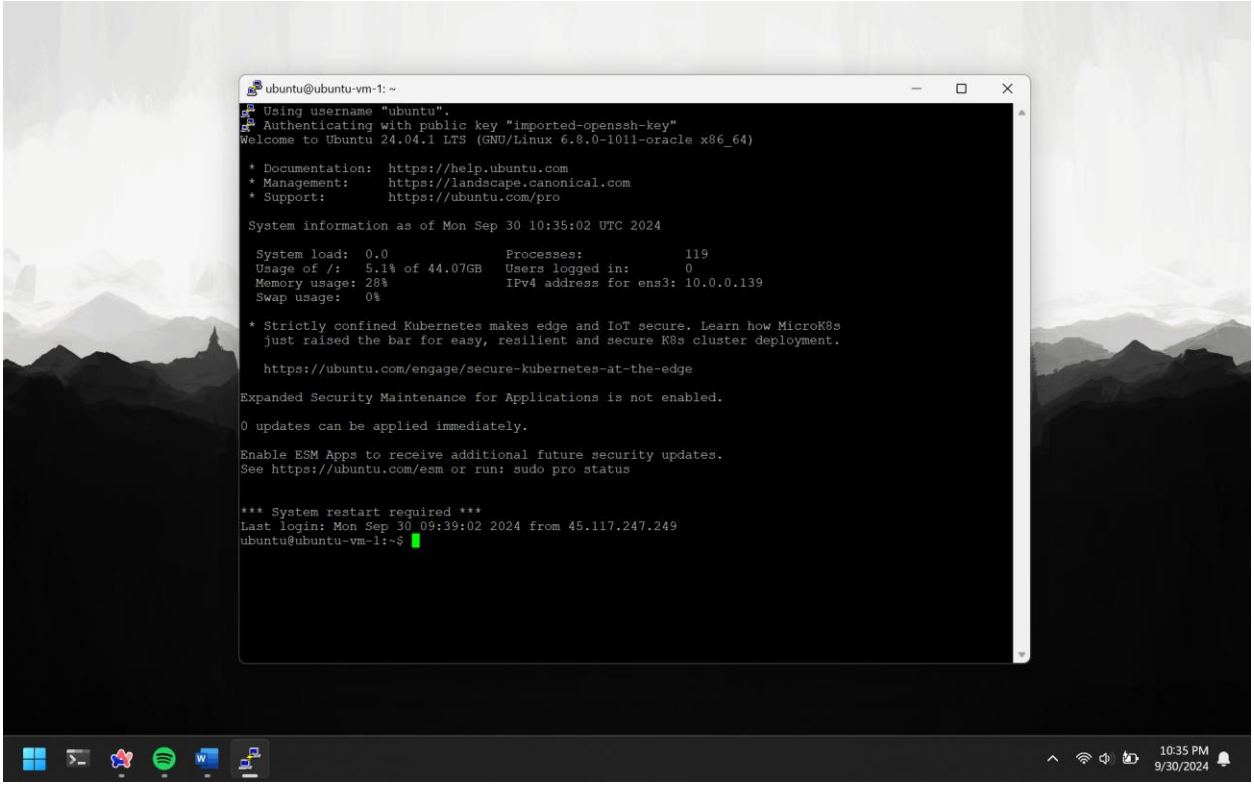
Step 9: On the left sidebar, click on “connection”, then “ssh”, then “auth”, and finally “credentials”. Click on “Browse” and select the new version of the private key that was obtained using PuTTYgen.



Step 10: On the left sidebar, click on “session” and then click “save” to save the current session configuration.



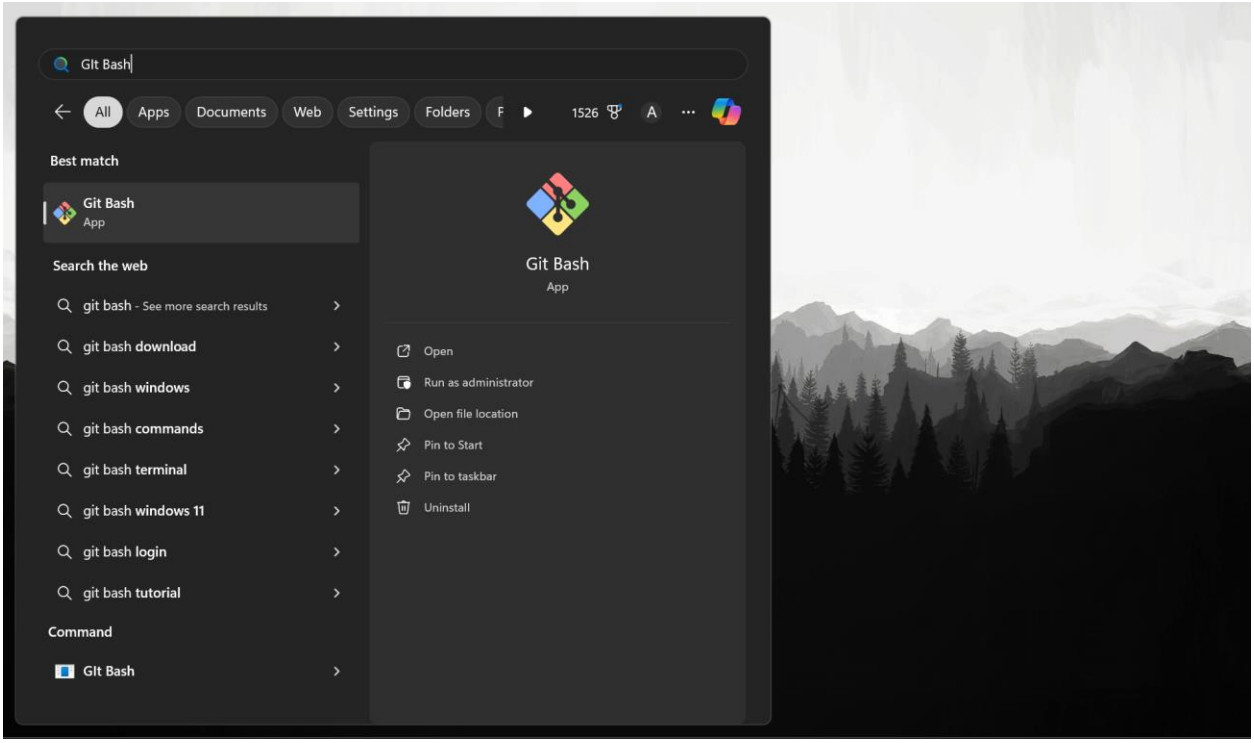
Step 11: Double click on the saved session name to access the Ubuntu VM.



Question 1: Access VM Using Git Bash

Step 1: Download and install Git. The Git Bash shell is included with Git.

Step 2: Search for the “Git Bash” application. Right click on it and click “Run as administrator”.

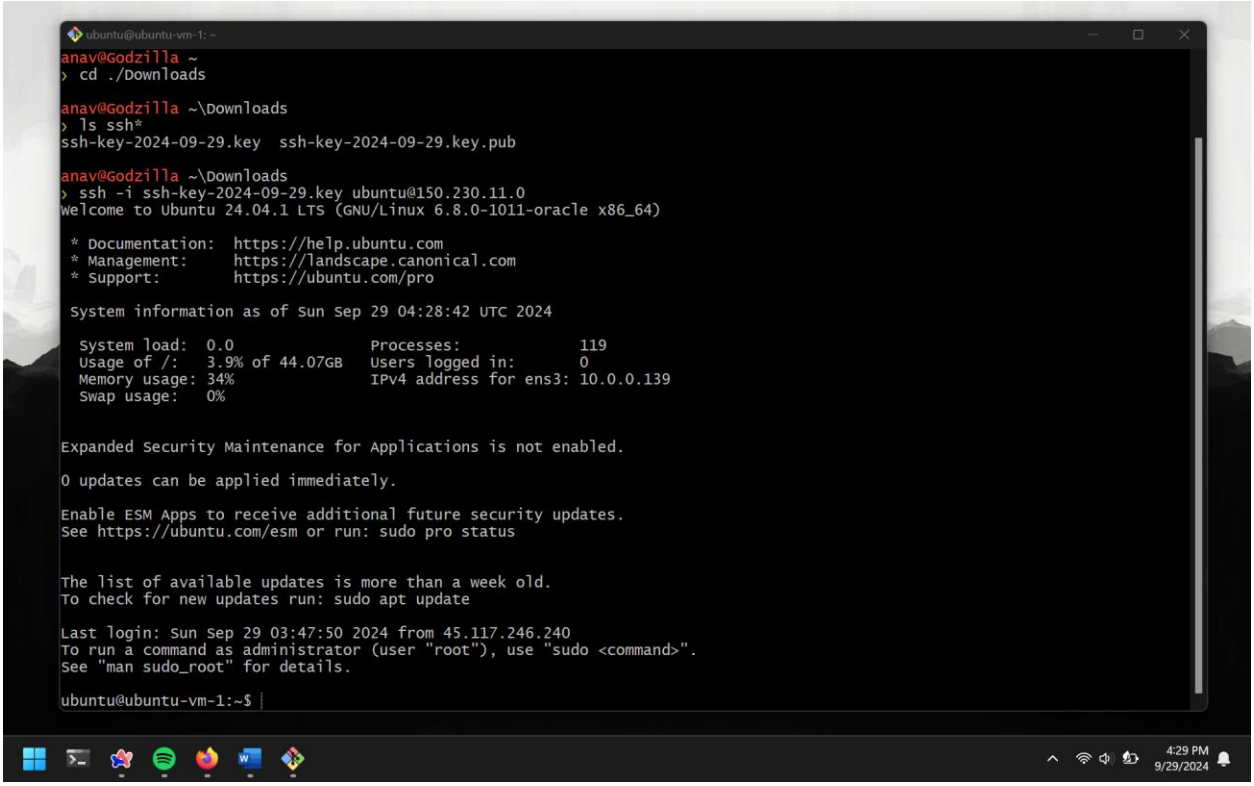


Step 3: Run the following command to access the machine:

ssh -i path-to-private-key hostname@public-ip-address

The path to the private key can be obtained my right clicking on the private key itself and selecting “Copy as path”.

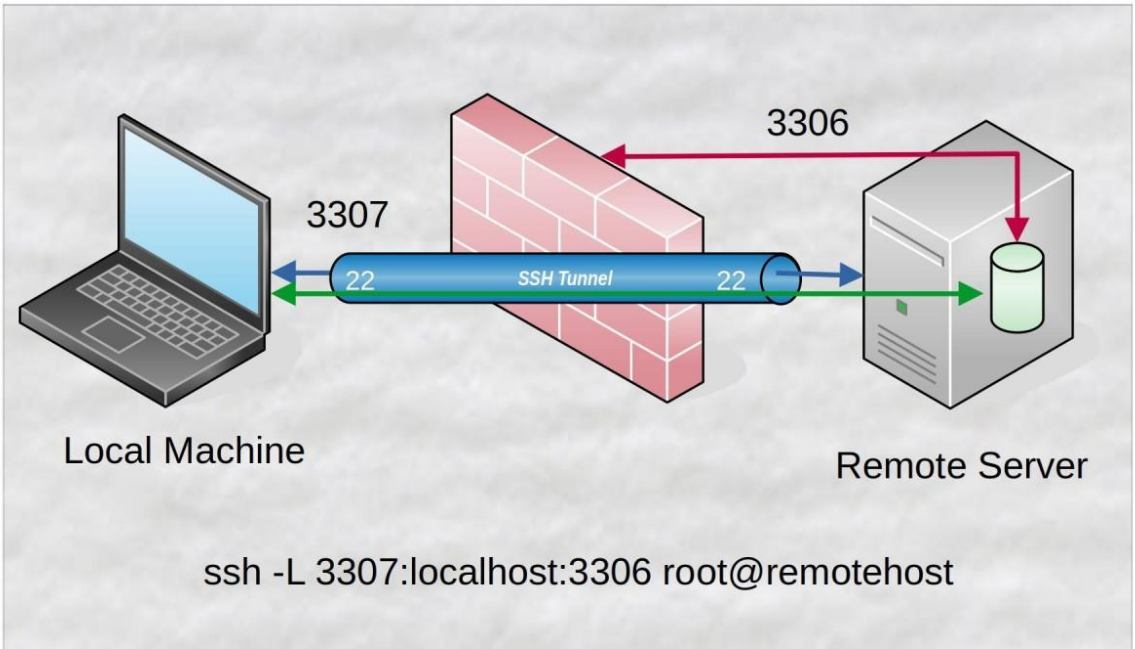
Both the hostname and public IP address can be found on Oracle Cloud. Go to https://cloud.oracle.com/dashboard, click on “instances” and select the desired VM to get its details.



ubuntu@ubuntu-vm-1 confirms the connection to the VM was successful.

Question 1: SSH and Telnet Comparison

Secure Shell (SSH) and Telnet are network protocols that are used to remotely access computers. SSH runs on port 22, while Telnet runs on port 23. During sessions that make use of remote access, SSH is preferred over telnet as it is more secure. This is a result of various features of SSH. The first of these security mechanisms is secure remote login (SSH Academy, 2023). This limits only authorized users to access a computer, for example using private and public keys to access virtual machine instances. Another security mechanism is remote encrypted connections (Cloudflare, 2024). This ensures data that is in transit between the SSH client and server is always encrypted. This encryption is achieved through algorithms such as Diffie-Hellman and RSA. These security mechanisms are what makes SSH a great choice for remote access, especially due to the fact that administrators usually have to remotely access networks and machines for critical tasks. Despite the lack of security in Telnet, it is still for various use cases. The first use case is to manage older systems (ClouDNS, 2024). Some older computer systems do not support SSH, which makes Telnet a good alternative to manage those systems. Another use case for Telnet is testing (ClouDNS, 2024). It can be used to check if connections can be established with servers, as well as check if ports are working as expected. For example, developers can use telnet to check if their server is accepting HTTP and HTTPS requests.



Question 2: Hello Cloud Website

Configure GUI Configure Apache	Detailed video guide
Configure PHP Create website Test localhost Test public IP	Detailed video guide

Public IP addresses:

- http://192.9.174.28
- http://152.67.126.88

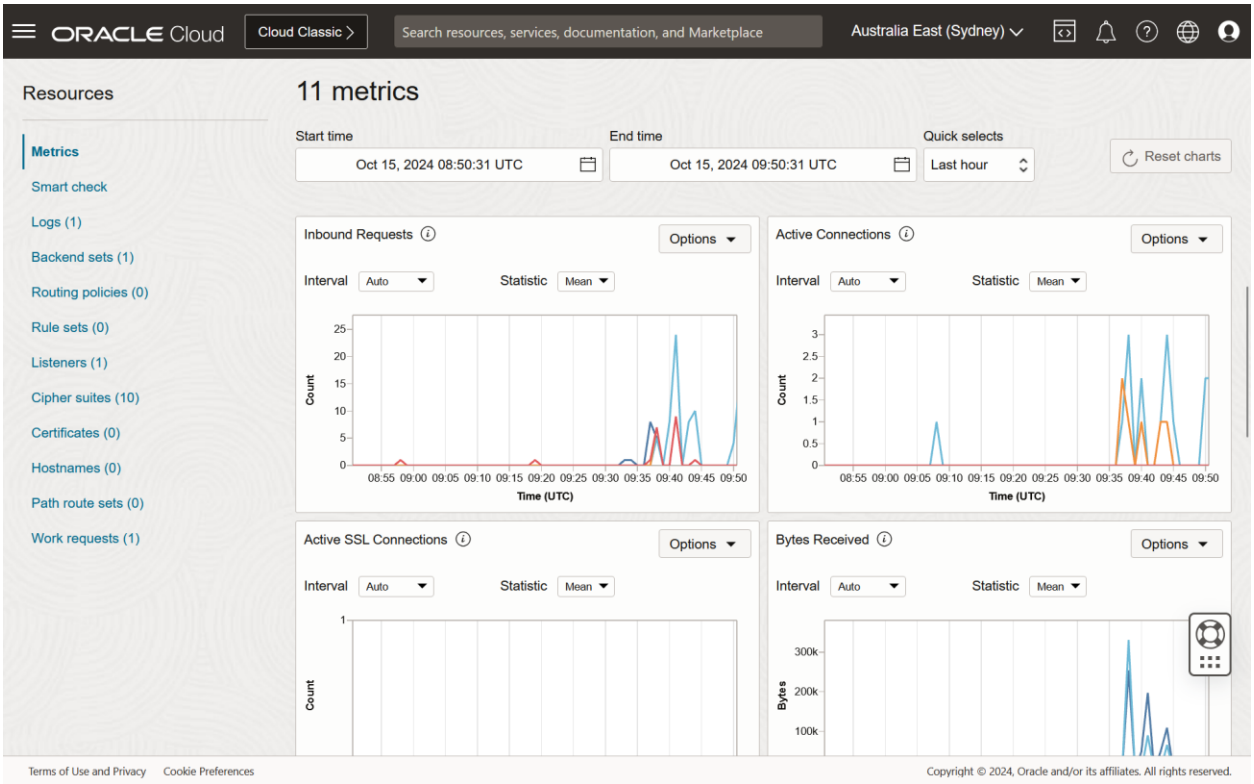
Question 3: WordPress Website

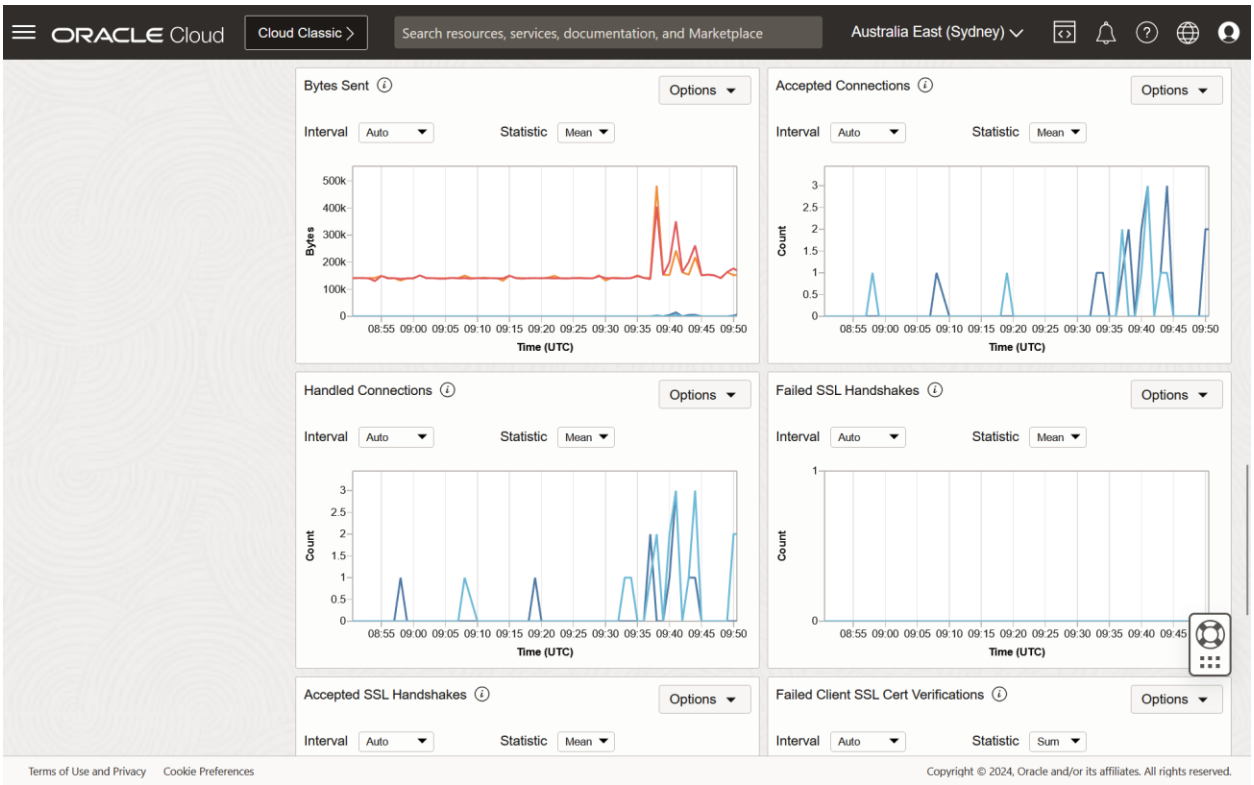
Configure WordPress	Detailed video guide
Configure template Generate content Access website	Detailed video guide
Configure load balancer Test load balancer	Detailed video guide

Public IP addresses:

- http://192.9.187.40 (Website)
- http://159.13.35.190 (Load balancer)

Load balancer metrics:




ORACLE Cloud
Cloud Classic
Search resources, services, documentation, and Marketplace
Australia East (Sydney)
Bytes Sent
Options
Interval
Auto
Statistic
Mean
Bytes
Time (UTC)
Accepted Connections
Count
Handled Connections
Failed SSL Handshakes
Accepted SSL Handshakes
Failed Client SSL Cert Verifications
Sum
Terms of Use and Privacy
Cookie Preferences
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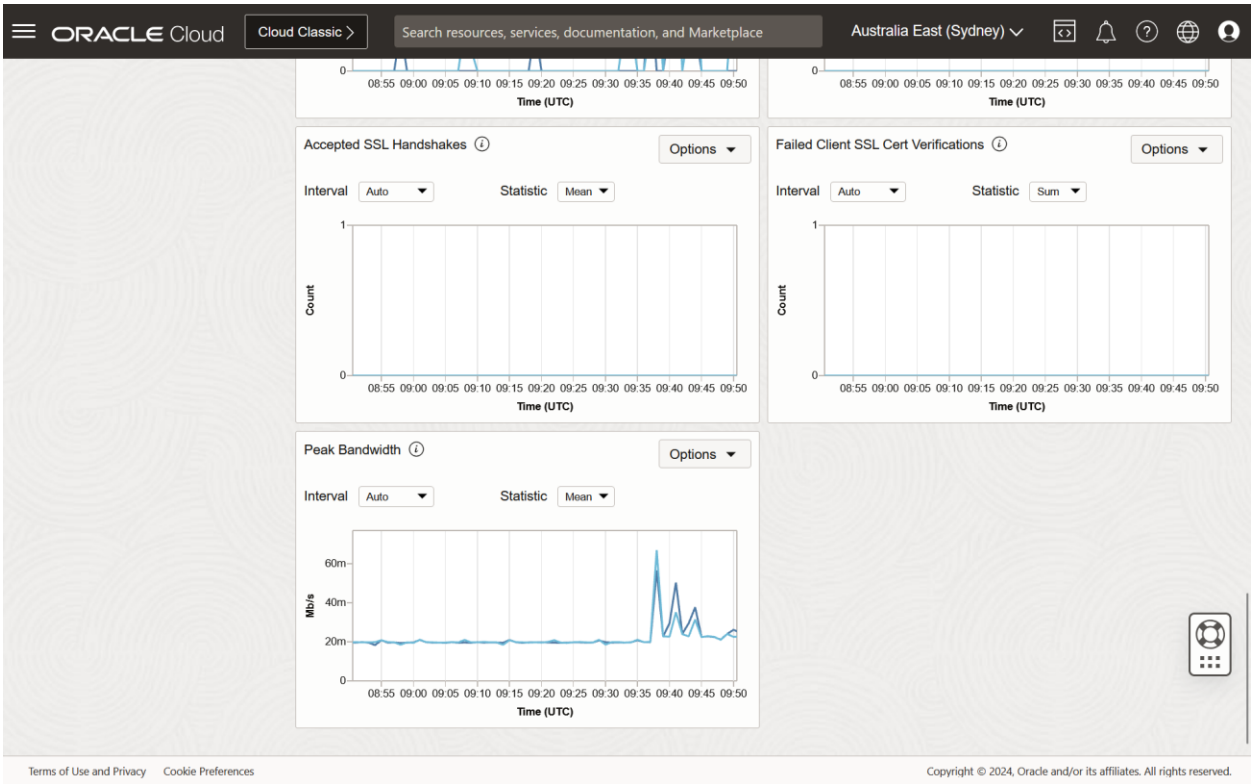
ORACLE Cloud
Cloud Classic
Search resources, services, documentation, and Marketplace
Australia East (Sydney)
Accepted SSL Handshakes
Options
Interval
Auto
Statistic
Mean
Count
Time (UTC)
Failed Client SSL Cert Verifications
Sum
Peak Bandwidth
Mb/s
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